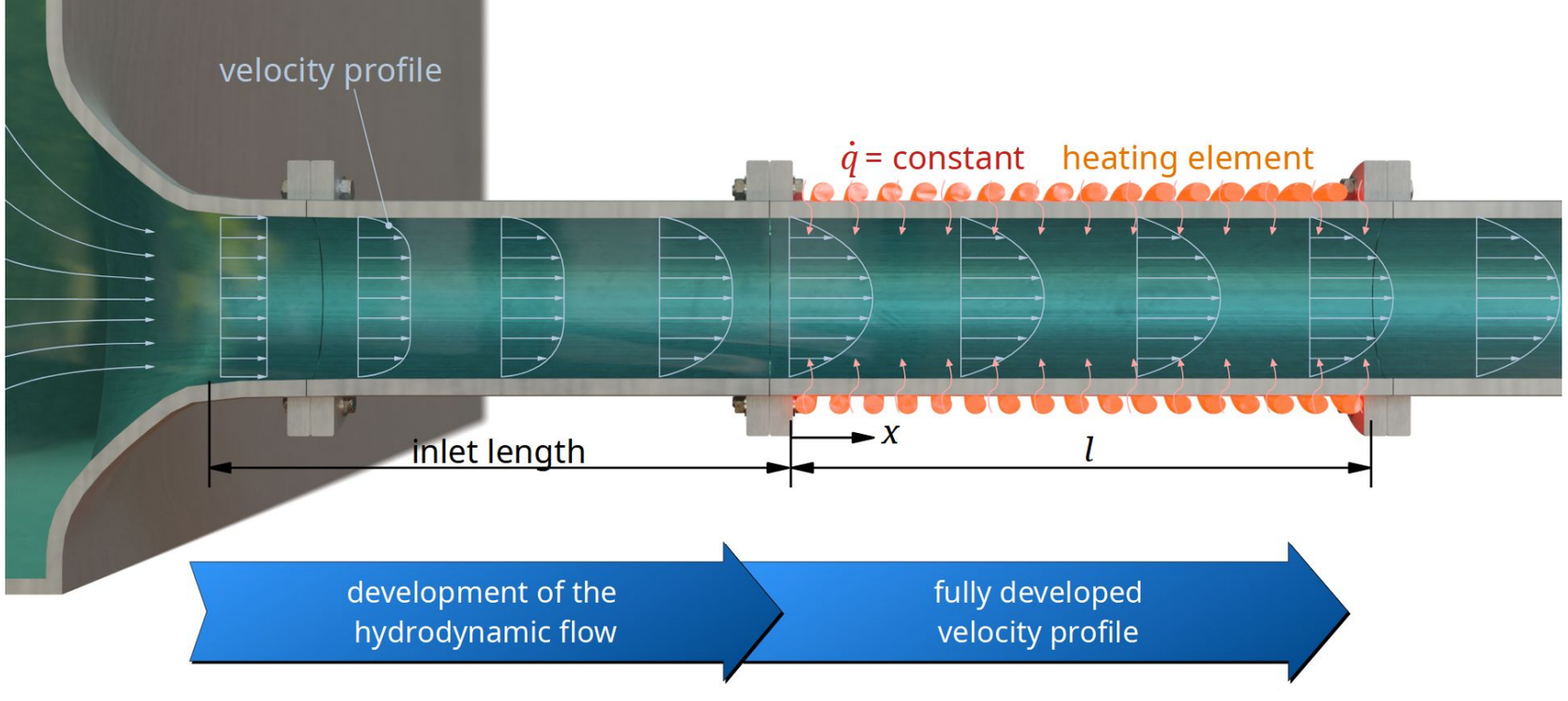
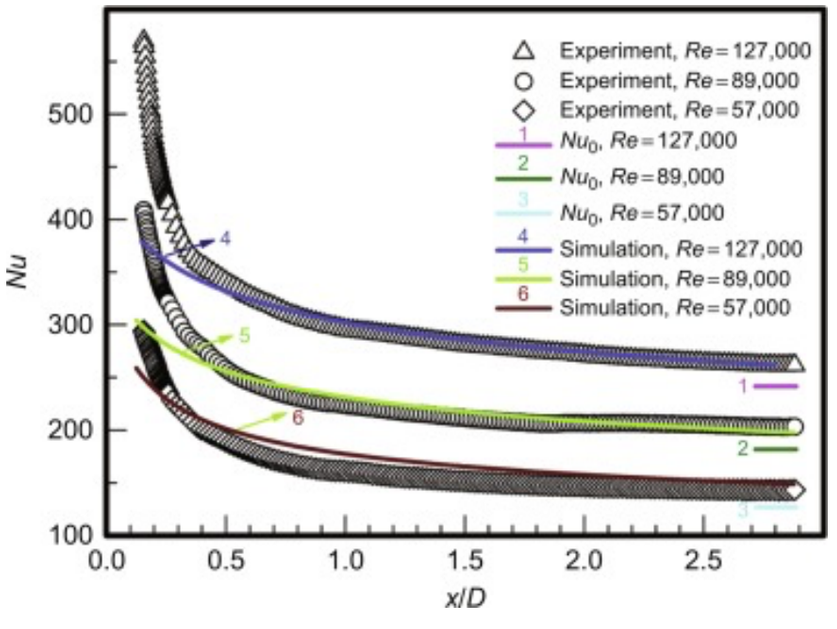
ENM 5310: Data-driven Modeling and Probabilistic Scientific Computing

Lecture #0: Introduction, motivation and course logistics

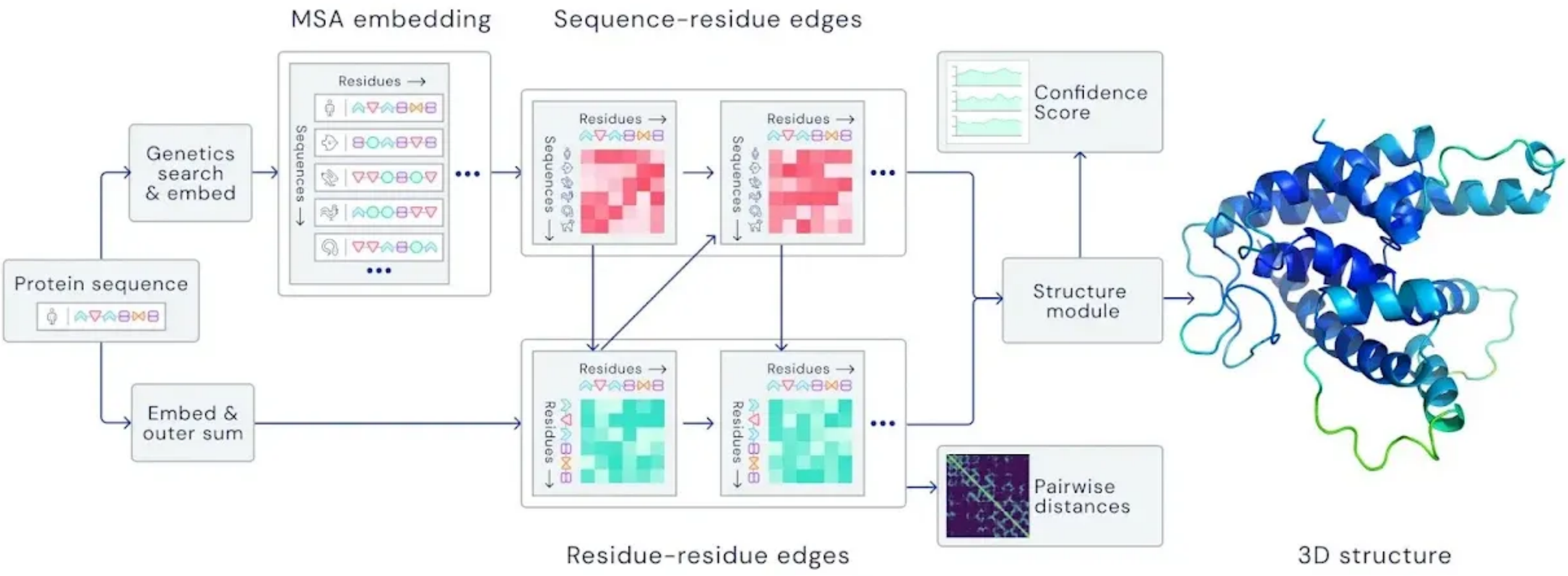




$$\mathrm{Nu} = 0.023 \cdot \mathrm{Re}^{0.8}\mathrm{Pr}^{0.4}$$

Dittus-Boelter equation: an empirical correlation used in fluid mechanics and heat transfer to calculate the Nusselt number (Nu) for turbulent fluid flow inside a pipe or tube.

- **Nu (Nusselt Number):** The ratio of convective heat transfer to conductive heat transfer at the fluid boundary. Solving for Nu allows you to calculate the actual heat transfer coefficient (h) of the system.
- **Re (Reynolds Number):** A dimensionless number that determines whether the fluid flow is laminar or turbulent by measuring the ratio of inertial forces to viscous forces.
- **Pr (Prandtl Number):** A dimensionless number representing the ratio of momentum diffusivity (viscosity) to thermal diffusivity.


MSA embedding
Sequence-residue edges
Residues
Sequences
Genetics search & embed
Protein sequence
Embed & outer sum
Residue-residue edges
Confidence Score
Structure module
Pairwise distances
3D structure

Data-driven Modeling

Data-driven modeling is the construction of mathematical models whose:

- *Structure chosen by us* — where physics and inductive bias enter (e.g. Dittus–Boelter's power-law form was a choice.
- *Content determined by data* — estimation (e.g. three coefficients)

ENM5310 is about:

- *How to build such models?* We will discuss the mathematical and computational underpinnings of modern machine learning.

- *How much to trust them?* This part most courses drop and this one will not. For example, Dittus–Boelter has a validity envelope, and nothing in the correlation tells you when you have left it.

I. predict	II. compress	III. decide
supervised	unsupervised	sequential / active

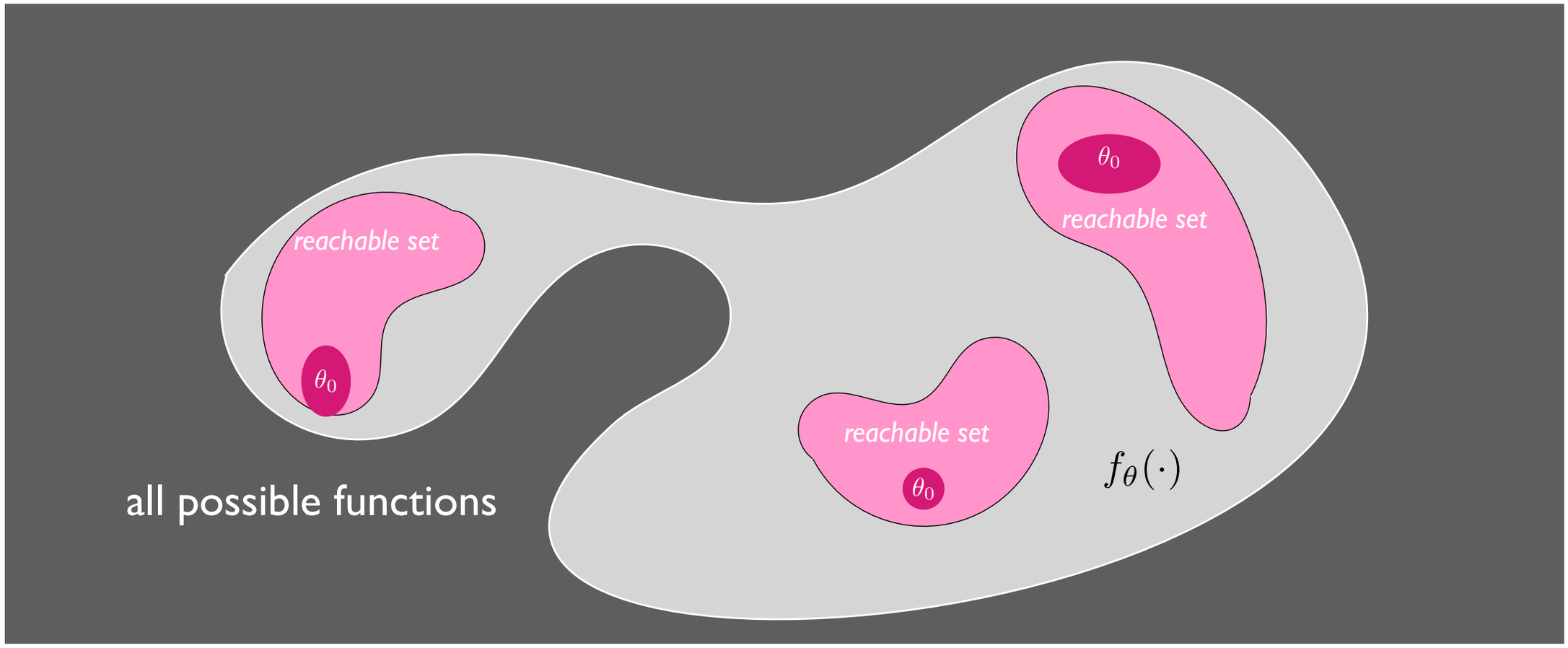
Everything on the syllabus is one of these three problems, plus the shared machinery for solving them. If at any point in the semester you cannot tell me which column we are in, stop me!

Supervised learning

Given $\{(x_i, y_i)\}_{i=1}^n \sim p(x, y)$, find $f : \mathcal{X} \to \mathcal{Y}$ minimizing

$$\mathbb{E}_{(x,y)\sim p}\big[\ell(y, f(x))\big]$$

Supervised learning



Key elements:

- Architecture/inductive bias
- Initialization
- Training data
- Normalization/Non-dimensionalization
- Learning objective
- Learning algorithm / Optimization

Supervised learning

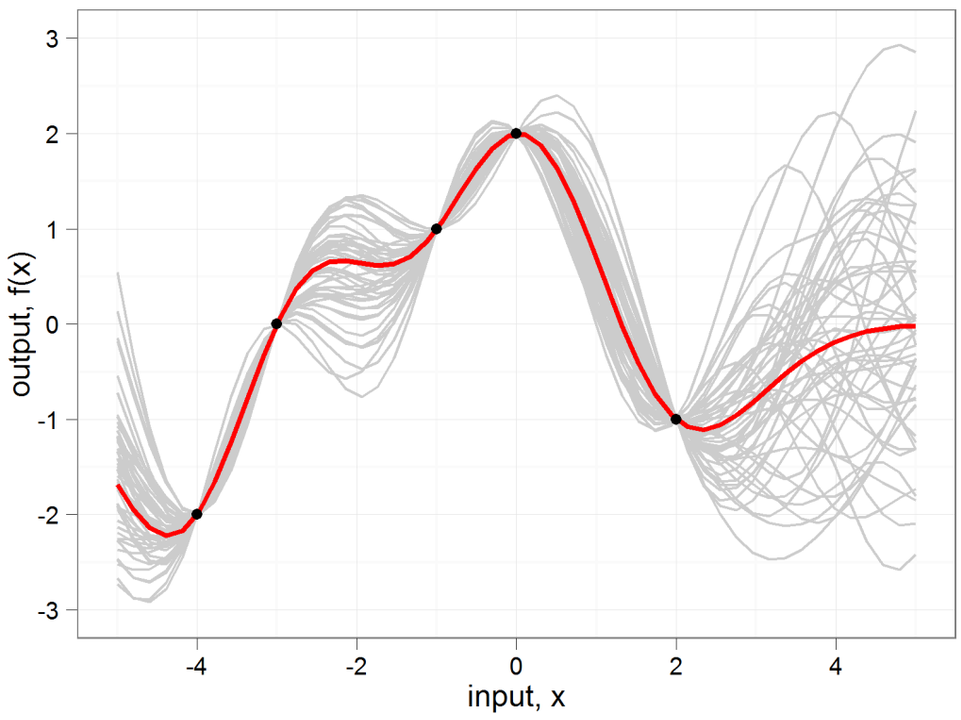
$$f : \mathcal{X} \to \mathcal{Y}$$

$$\mathcal{D} = \{\boldsymbol{x}, \boldsymbol{y}\},\ \boldsymbol{x} \in \mathcal{X},\ \boldsymbol{y} \in \mathcal{Y}$$

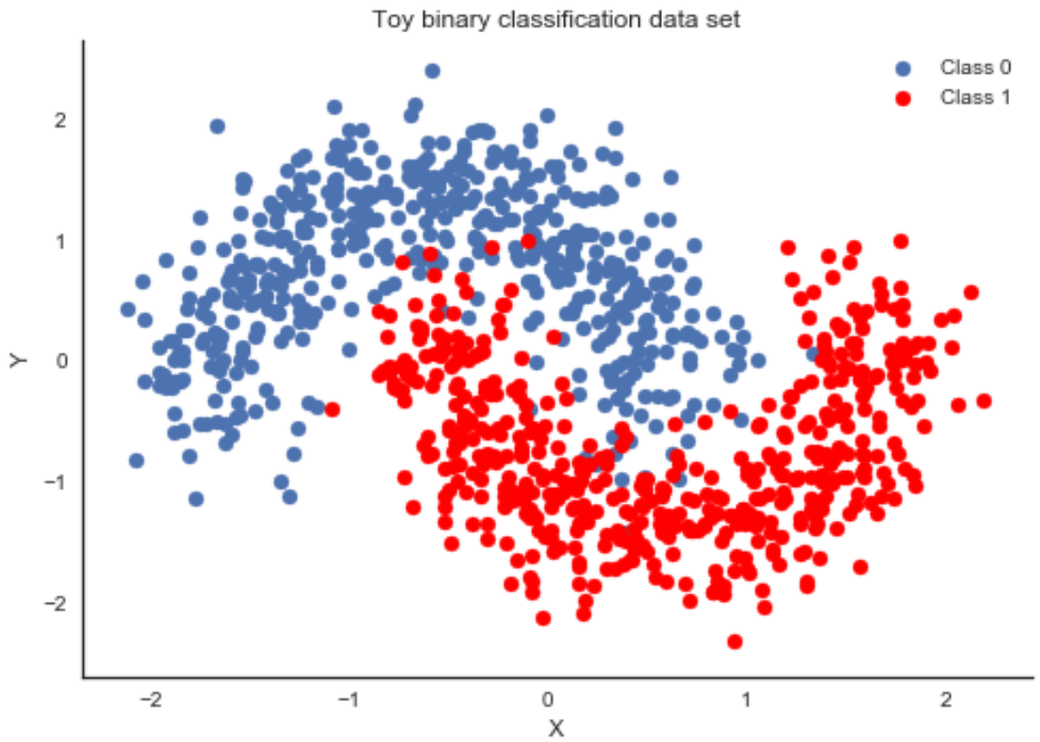
$$\boldsymbol{y} = f(\boldsymbol{x}) + \epsilon$$

$$p(f(\boldsymbol{x}^*)|\boldsymbol{x}^*, \mathcal{D})$$

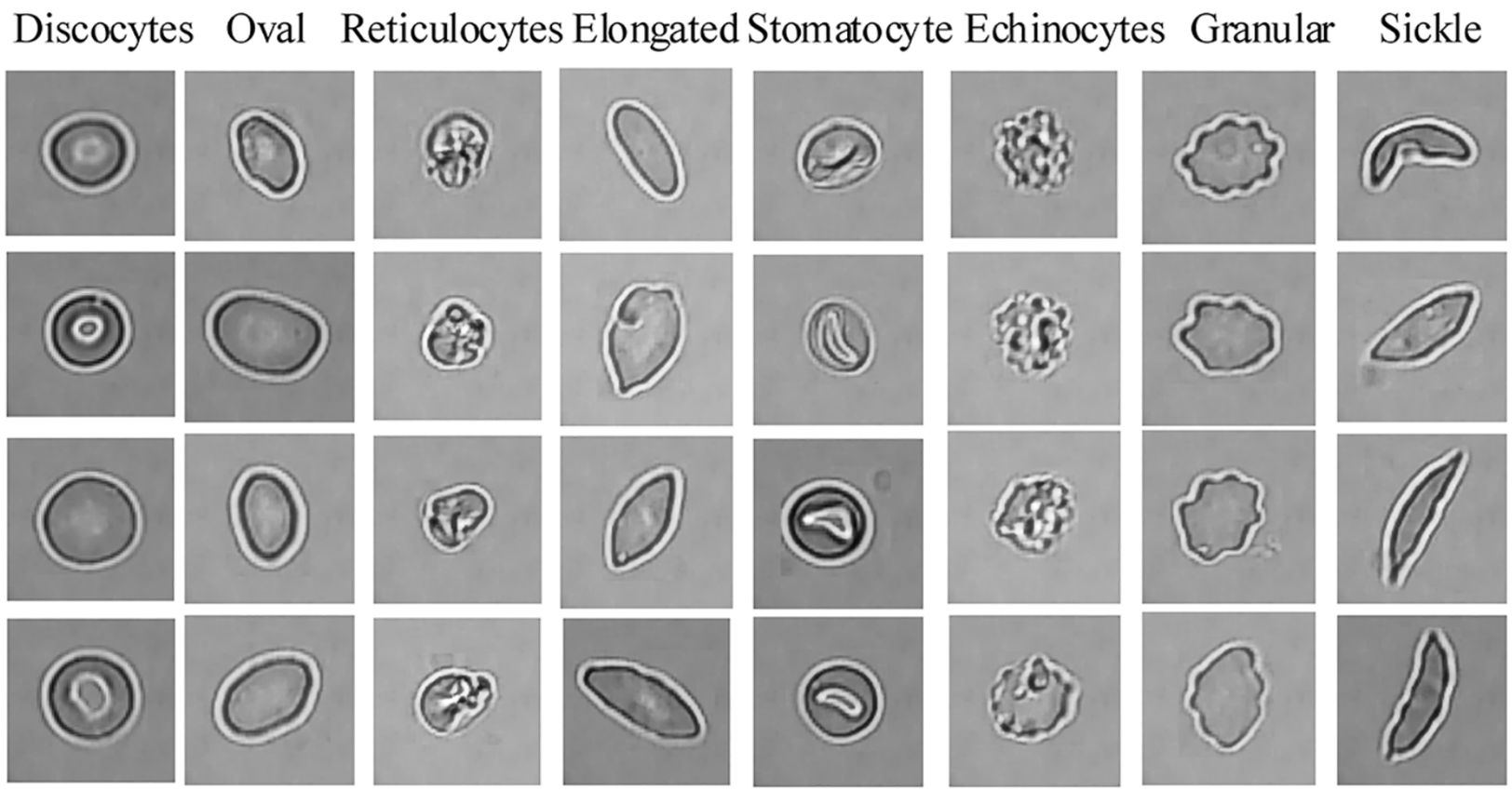
Regression



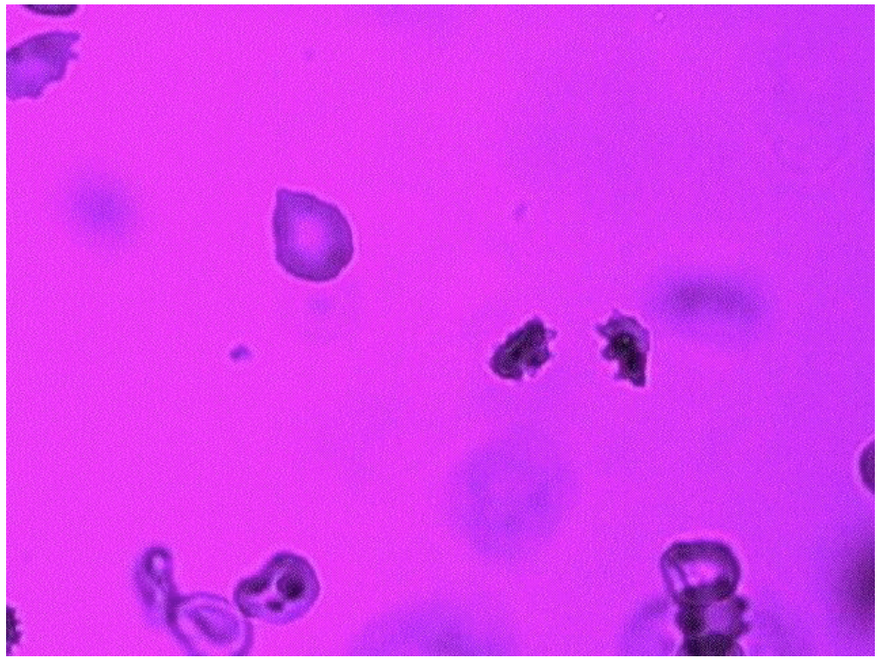
Classification



Classification

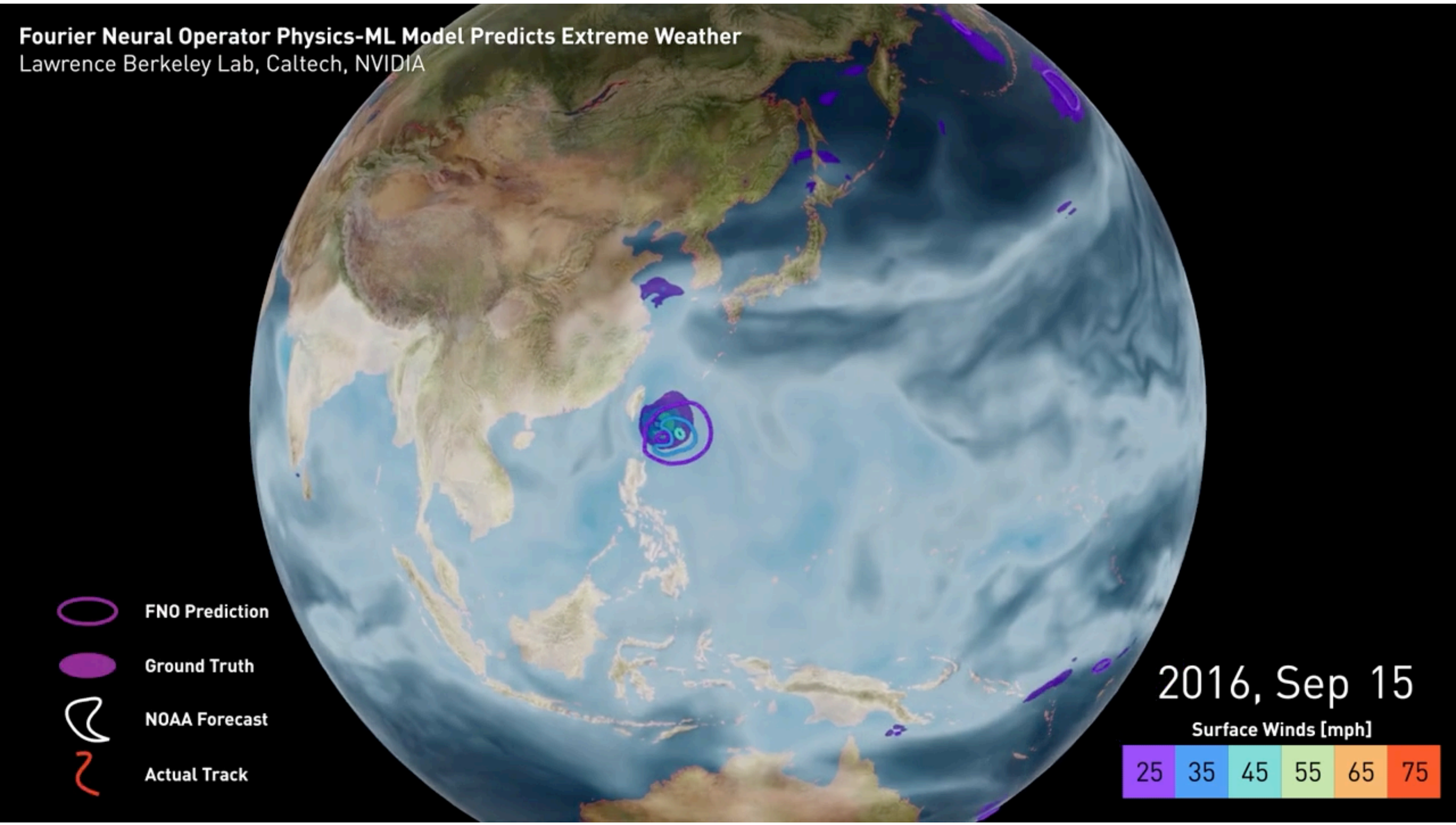


$$\mathcal{D} = \{\boldsymbol{x}, \boldsymbol{y}\},\ \boldsymbol{x} \in \mathcal{X},\ \boldsymbol{y} \in \mathcal{Y}$$



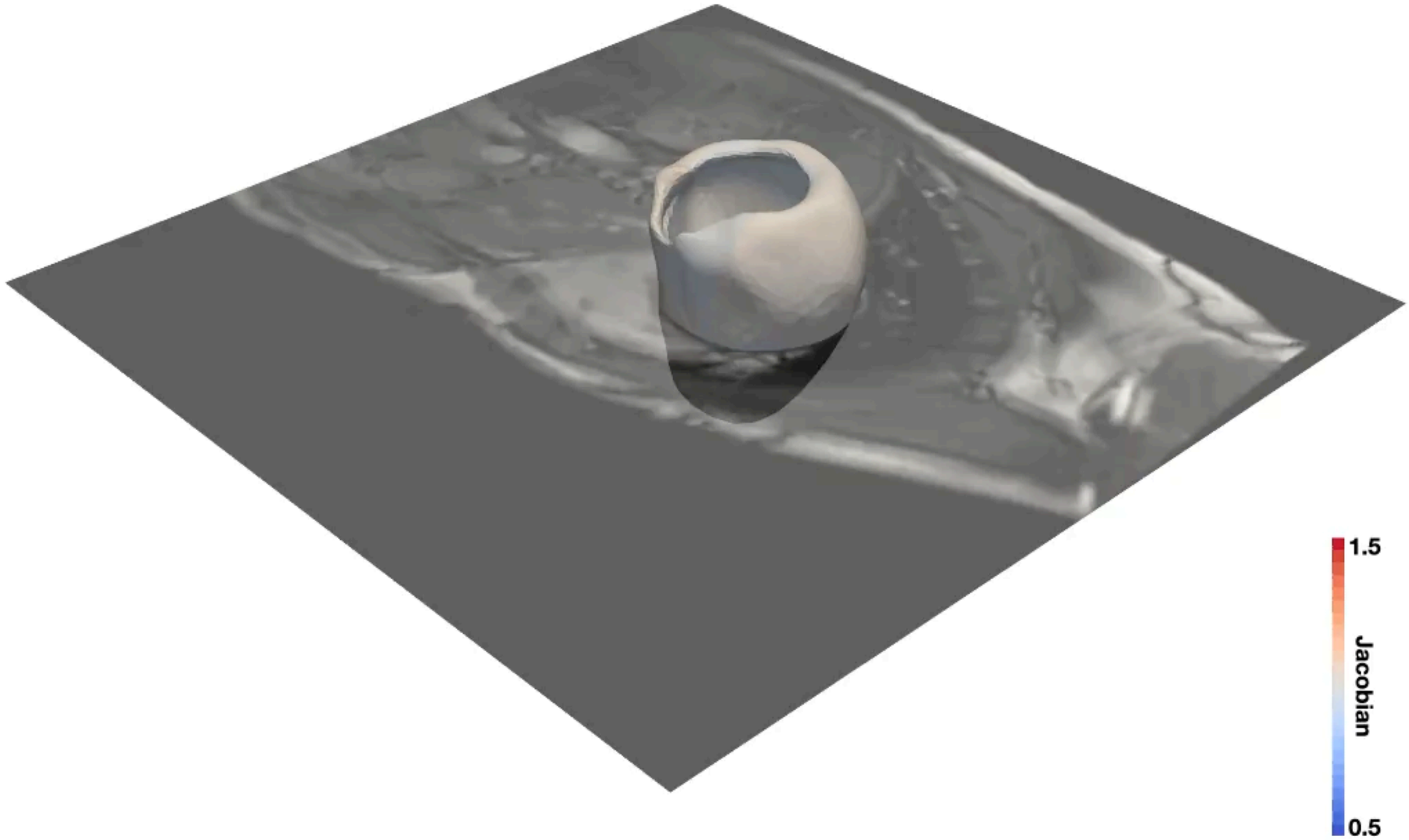
$$\boldsymbol{y} = f(\boldsymbol{x}) + \epsilon$$

Spatio-temporal prediction



https://blogs.nvidia.com/blog/2021/11/16/ai-science-climate-change/

Self-supervised Learning

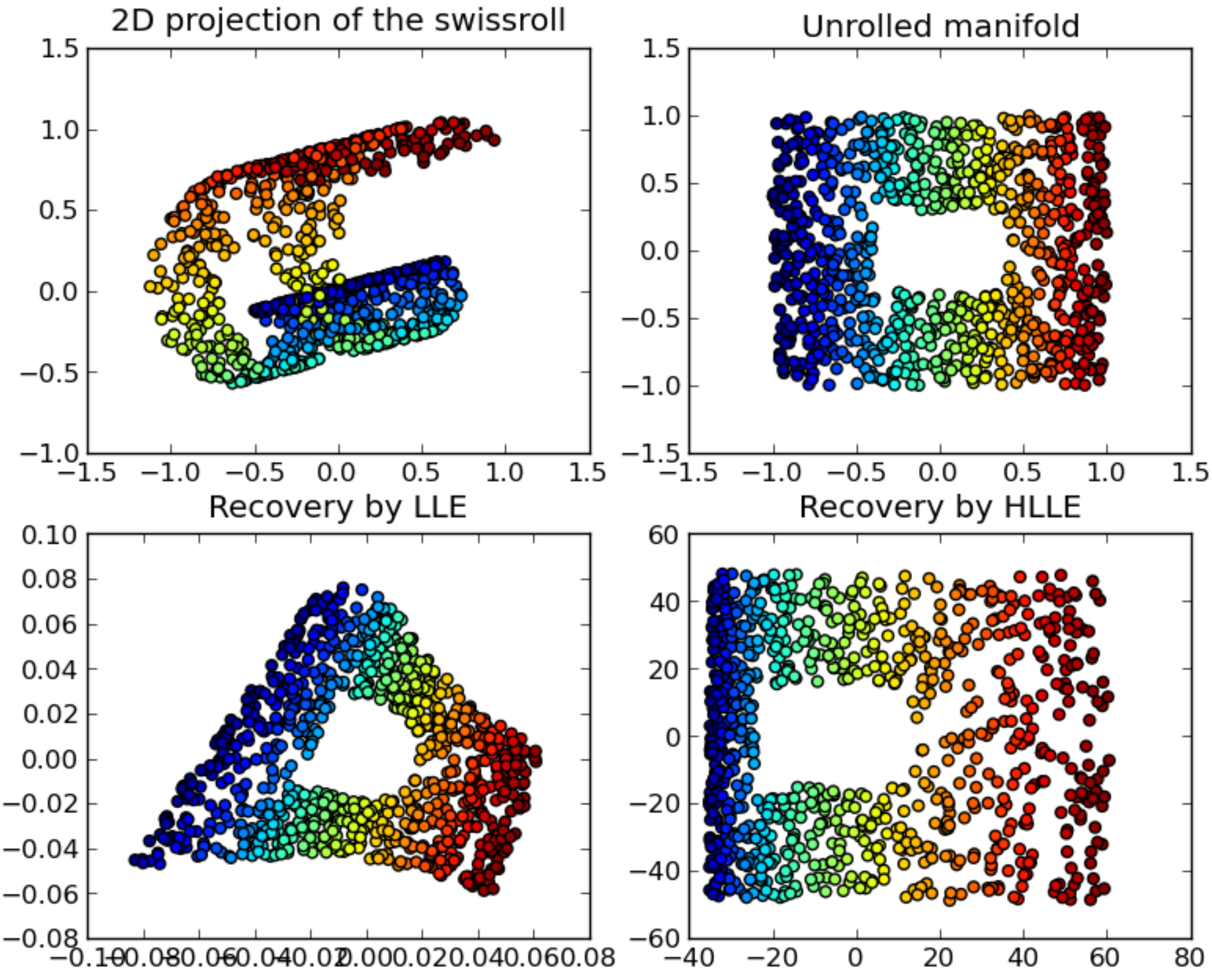


Unsupervised Learning

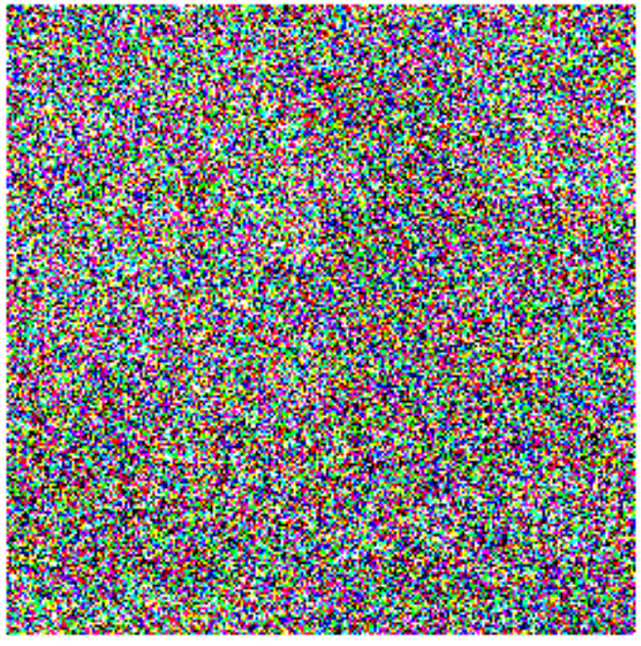
$$\text{Given } \{x_i\}_{i=1}^n, \text{ find } e : \mathcal{X} \to \mathcal{Z} \text{ and } d : \mathcal{Z} \to \mathcal{X}$$
$$\text{with } \dim \mathcal{Z} \ll \dim \mathcal{X} \text{ and } d(e(x)) \approx x$$

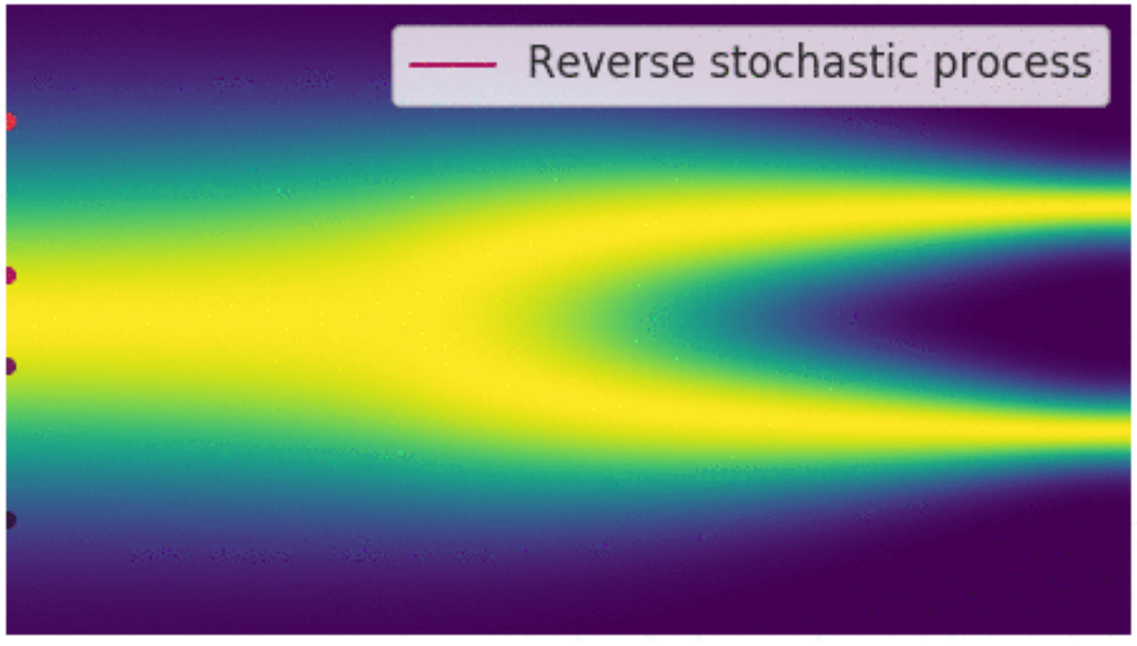
or, more honestly: find a $p(x)$ you can *sample* from

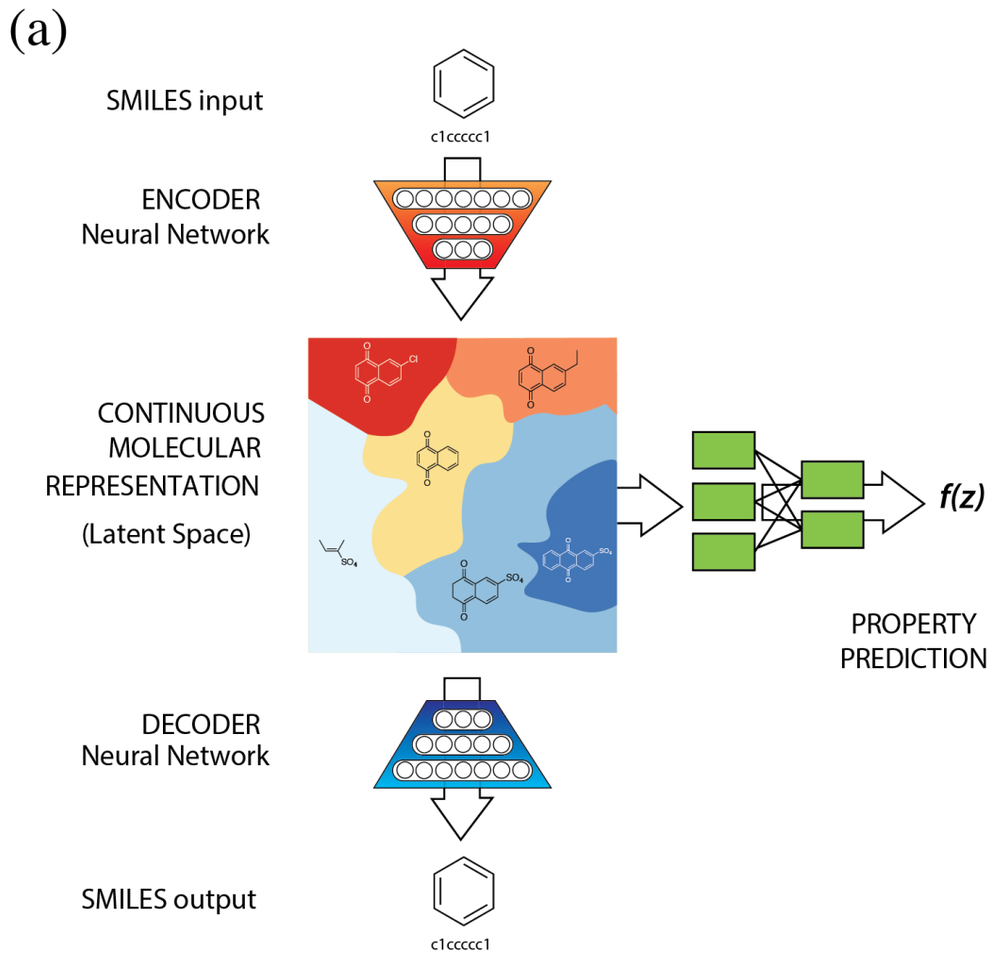
Dimensionality reduction

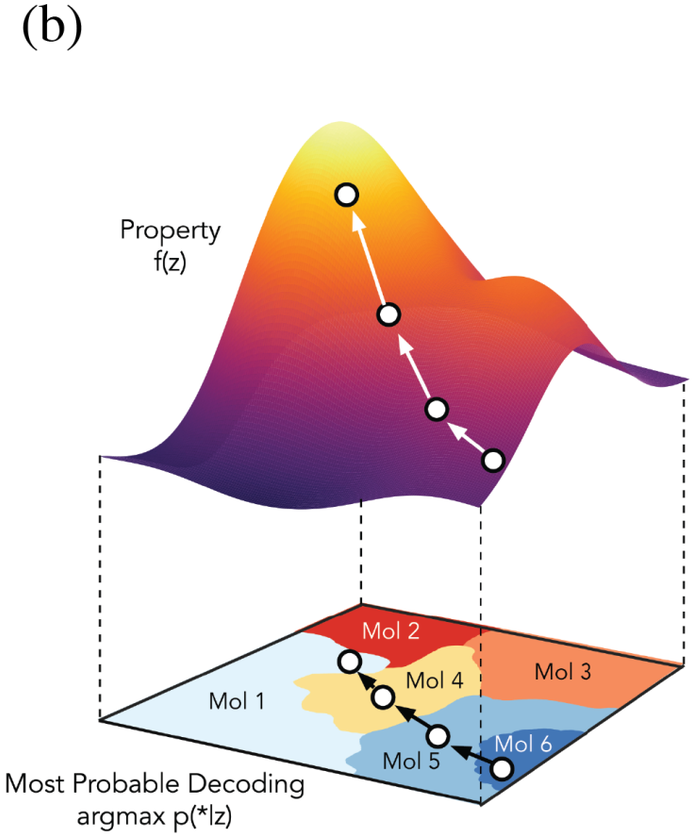


Generative Models







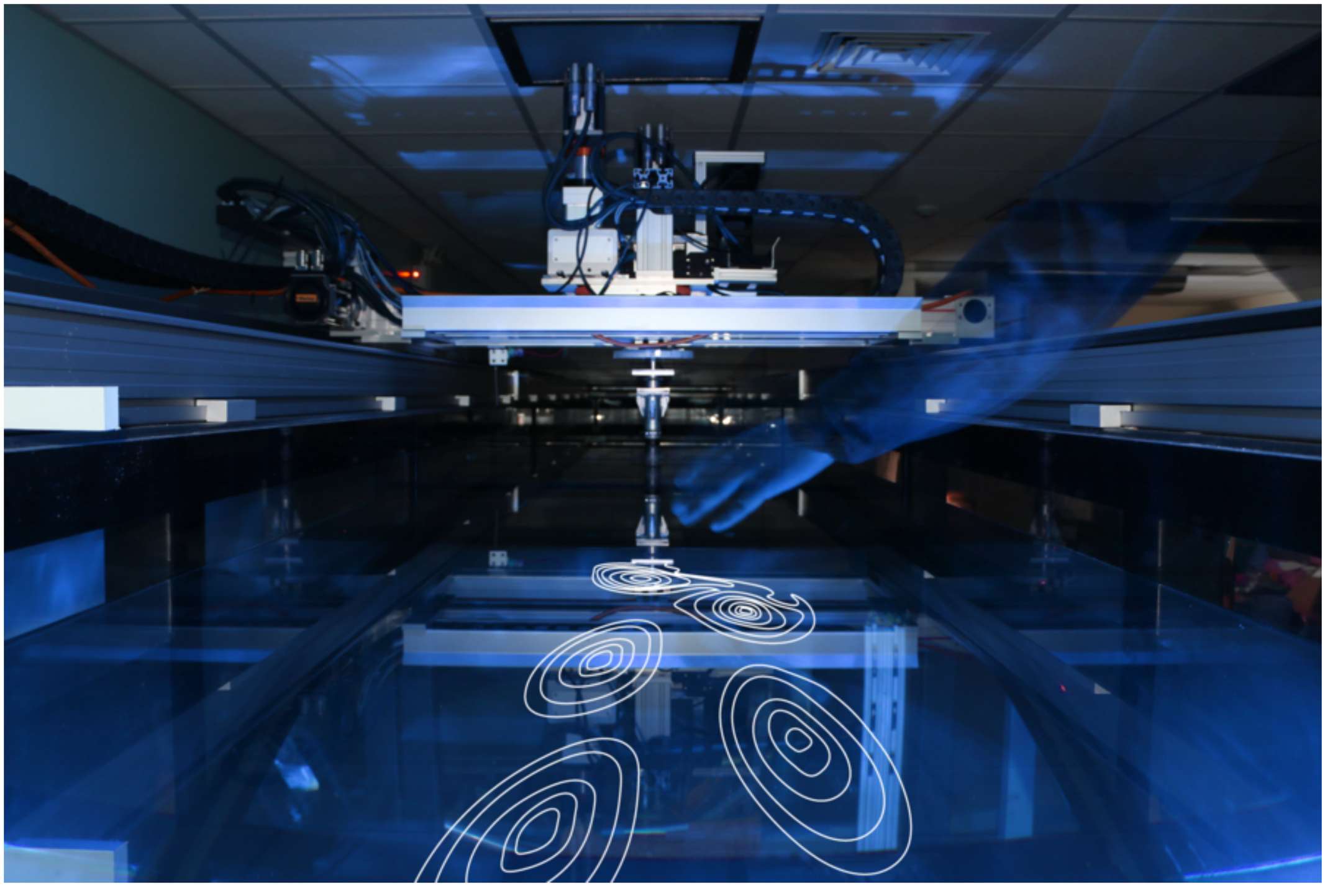


Sequential Decision Making

$$x_{n+1} = \arg\max_{x \in \mathcal{X}} \; \alpha\big(x \,;\, p(f \mid \mathcal{D}_n)\big)$$

the acquisition depends on the posterior distribution, not on a point estimate.
No uncertainty, no decision-making!

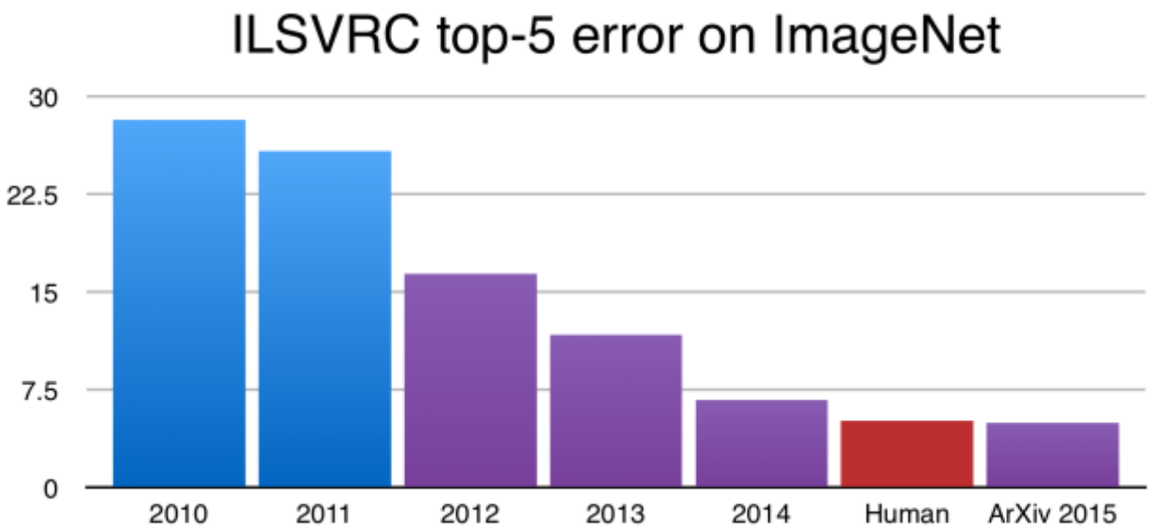
Intelligent experimentation

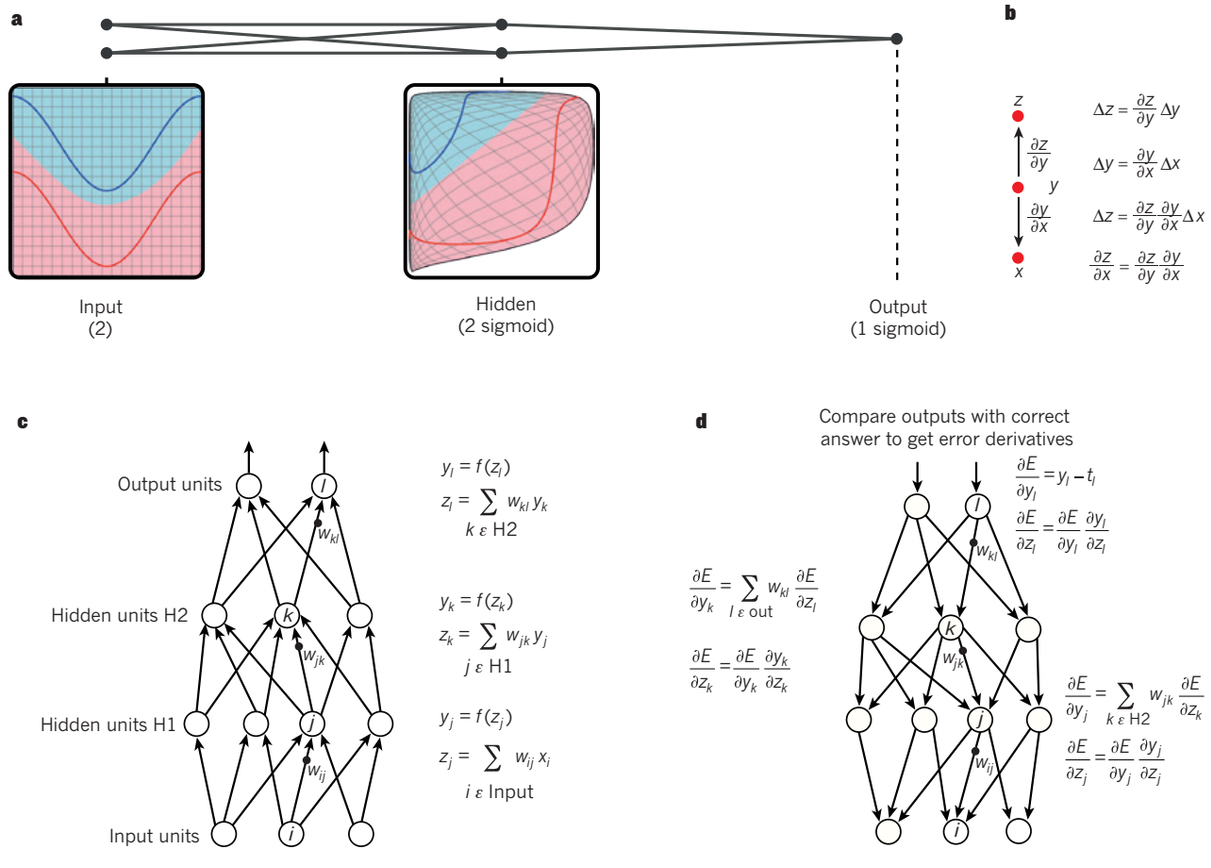


"In its first year of operation, the Intelligent Towing Tank (ITT) conducted about 100,000 total experiments, essentially completing the equivalent of a PhD student's five years' worth of experiments in a matter of weeks."

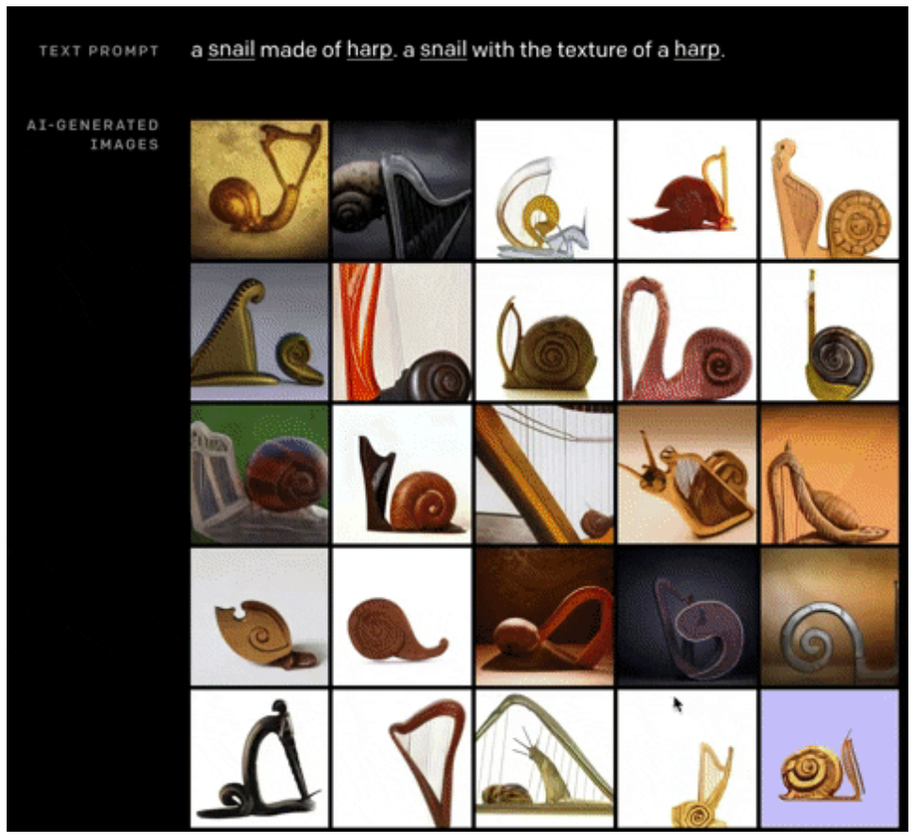
https://news.mit.edu/2019/intelligent-towing-tank-propels-research-1209

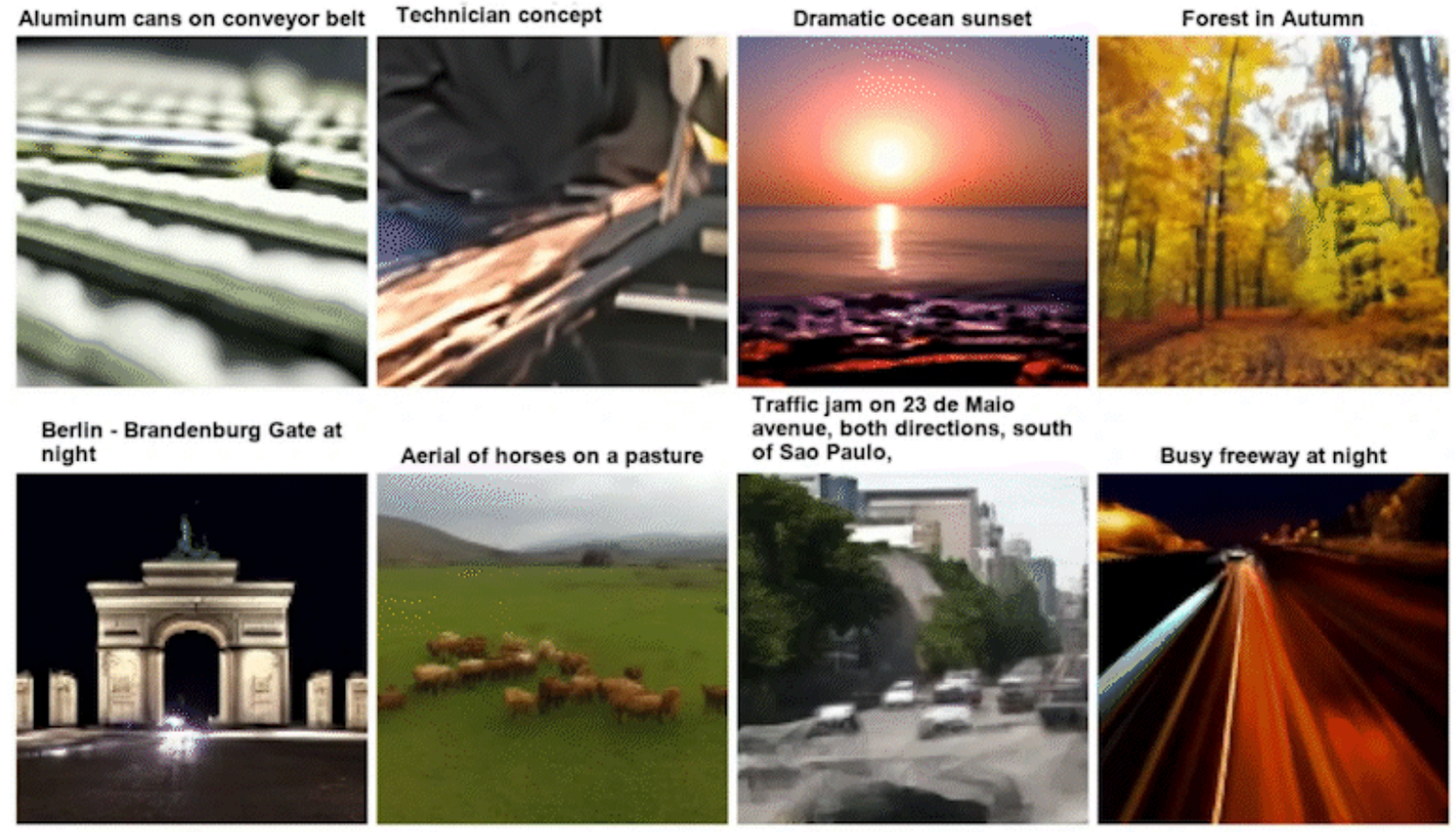
The Deep Learning Revolution

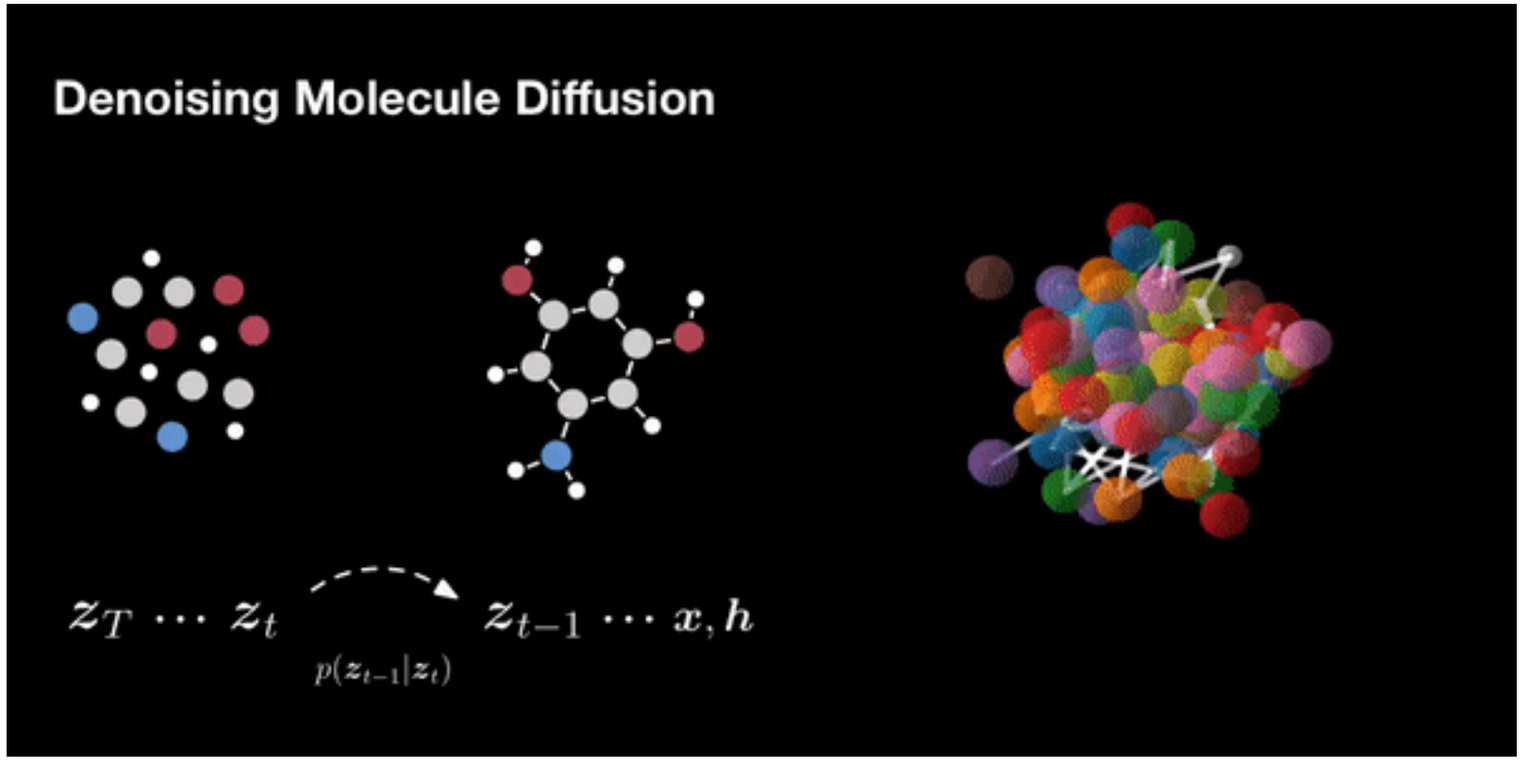




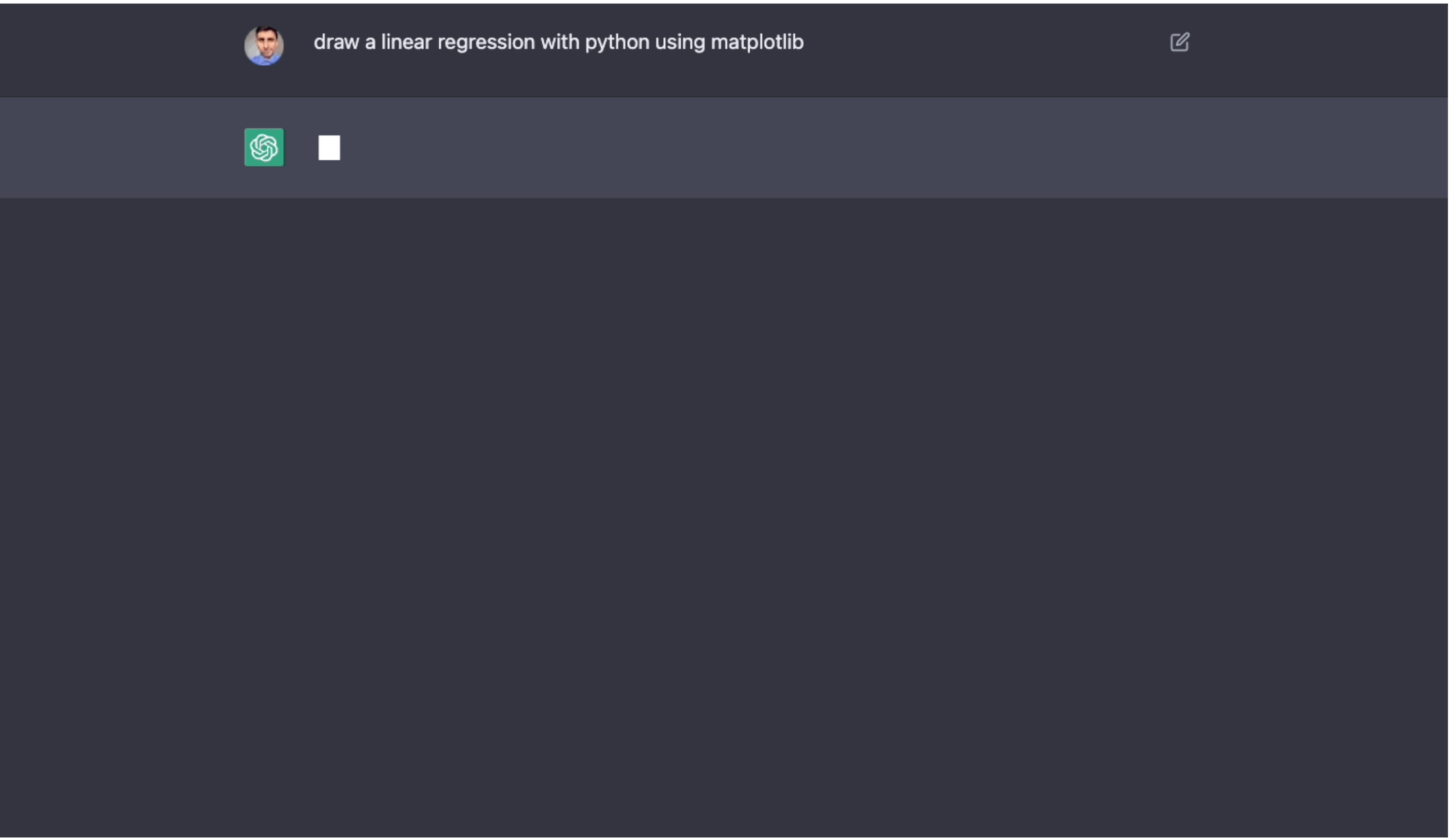
Working with high-dimensional data







Working with multiple data modalities



Is this the right course for me?

Yes, if you:

(a) Fulfill the pre-requisites (linear algebra, probability & statistics, optimization, Python programming).

(b) Want to build foundational knowledge in ML for engineering research and applications.

(c) Want to sharpen your implementation skills.

No, if you:

(a) Do not fulfill the pre-requisites.

(b) Are solely interested in theory.

(c) Are solely interested in CS applications (e.g. computer vision, NLP).

(d) Are solely interested in specialized topics (e.g. RL, geometric deep learning).

Course goals

Foundational Competencies:

- Master probabilistic modeling frameworks and Bayesian inference for scientific applications
- Implement advanced optimization algorithms including stochastic methods and second-order techniques
- Apply variational inference and MCMC sampling for complex probabilistic models
- Quantify and propagate uncertainty in computational models and data-driven predictions

Deep Learning Expertise:

- Design and implement neural networks for scientific computing applications
- Apply convolutional architectures to spatial data in engineering and physics
- Leverage Transformer models and attention mechanisms for scientific sequence modeling
- Build recurrent architectures for time-series analysis and dynamical system modeling

Physics-Informed AI:

- Develop physics-informed neural networks (PINNs) that respect physical constraints
- Implement neural operators for solving partial differential equations
- Integrate domain knowledge and conservation laws into machine learning models
- Apply Gaussian processes for multi-fidelity modeling and active learning

Generative Modeling:

- Build and train variational autoencoders for dimensionality reduction and data generation
- Implement diffusion models for high-quality sample generation
- Apply flow matching and normalizing flows for density estimation and sampling
- Design generative models for scientific data synthesis and augmentation

Systems Integration:

- Combine multiple ML techniques into coherent scientific computing pipelines
- Perform experimental design and active learning for efficient data collection
- Bridge traditional computational methods with modern AI approaches
- Deploy uncertainty-quantified ML systems for real-world engineering applications

Course logistics

- Duration: 15 Weeks
- Time: Tuesdays and Thursdays, 3:30pm to 5:00pm (AGH 105)
- Dates: August 25 to December 7, 2026
- Instructor office hours: Thursdays 12-2pm
- TA office hours: TBD
- Bi-weekly quizzes, 2 coding assignments, 2 midterms, final project

For more details visit the course website:
https://www.seas.upenn.edu/~enm5310/

Slides and code can be found on Github:
https://github.com/PredictiveIntelligenceLab/ENM5310

Tentative Syllabus

Part I: Foundations of Probabilistic Learning

Week 1: Probability theory, statistics, and information theory fundamentals
Week 2: Statistical estimation: Maximum likelihood, MAP, and Bayesian inference
Week 3: Optimization foundations: Gradients, Hessians, and stochastic optimization
Week 4: Linear models and generalized linear regression with uncertainty quantification

Part II: Advanced Inference and Sampling

Week 5: Variational inference and expectation maximization
Week 6: Markov Chain Monte Carlo and advanced sampling methods
Week 7: Midterm Exam

Part III: Deep Learning Architectures

Week 8: Multi-layer perceptrons and universal approximation theory
Week 9: Convolutional neural networks and their applications to scientific data
Week 10: Transformers and attention mechanisms for scientific computing
Week 11: Recurrent networks, LSTMs, and sequence modeling for time-series

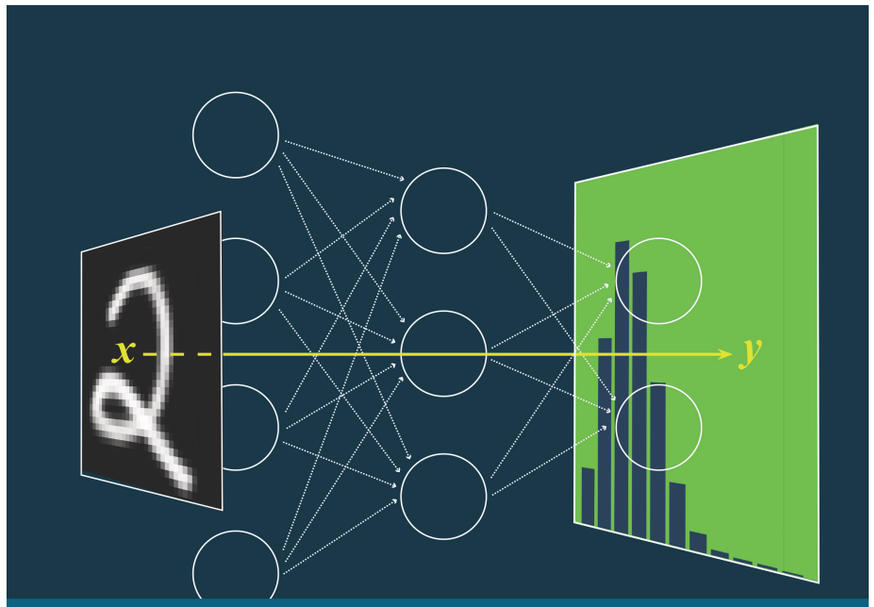
Part IV: Physics-Informed and Scientific Machine Learning

Week 12: Physics-informed neural networks (PINNs) and neural operators
Week 13: Kernel methods, Gaussian processes, and scientific applications

Part V: Generative Models and Advanced Topics

Week 14: Variational autoencoders, diffusion models, and flow matching
Week 15: Final project presentations and experimental design for ML

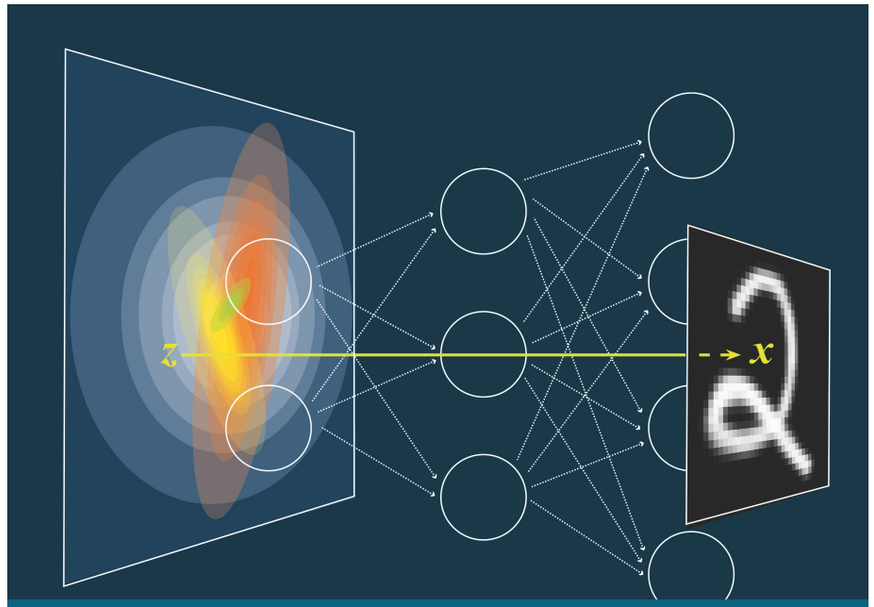
Textbooks



Probabilistic Machine Learning

An Introduction

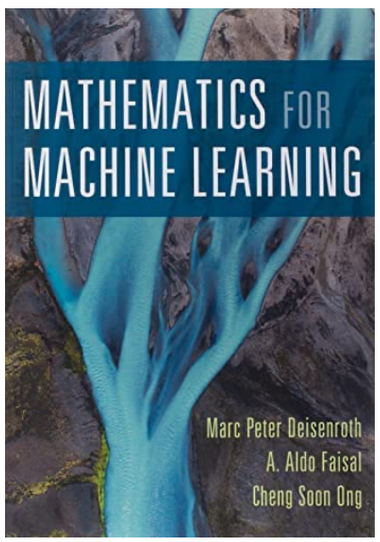
Kevin P. Murphy

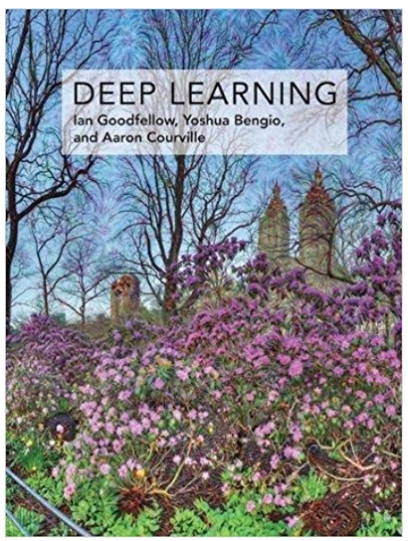


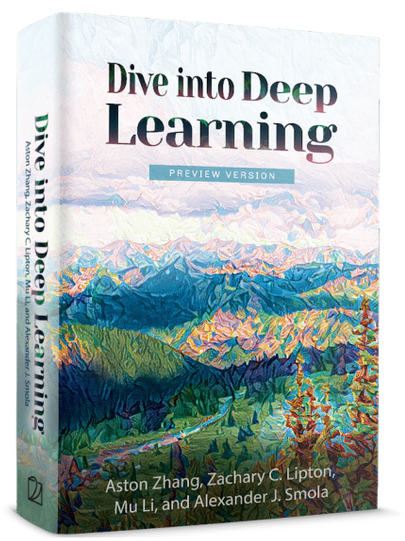
Probabilistic Machine Learning

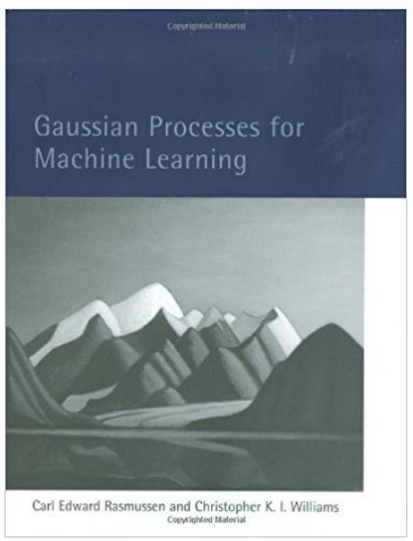
Advanced Topics

Kevin P. Murphy

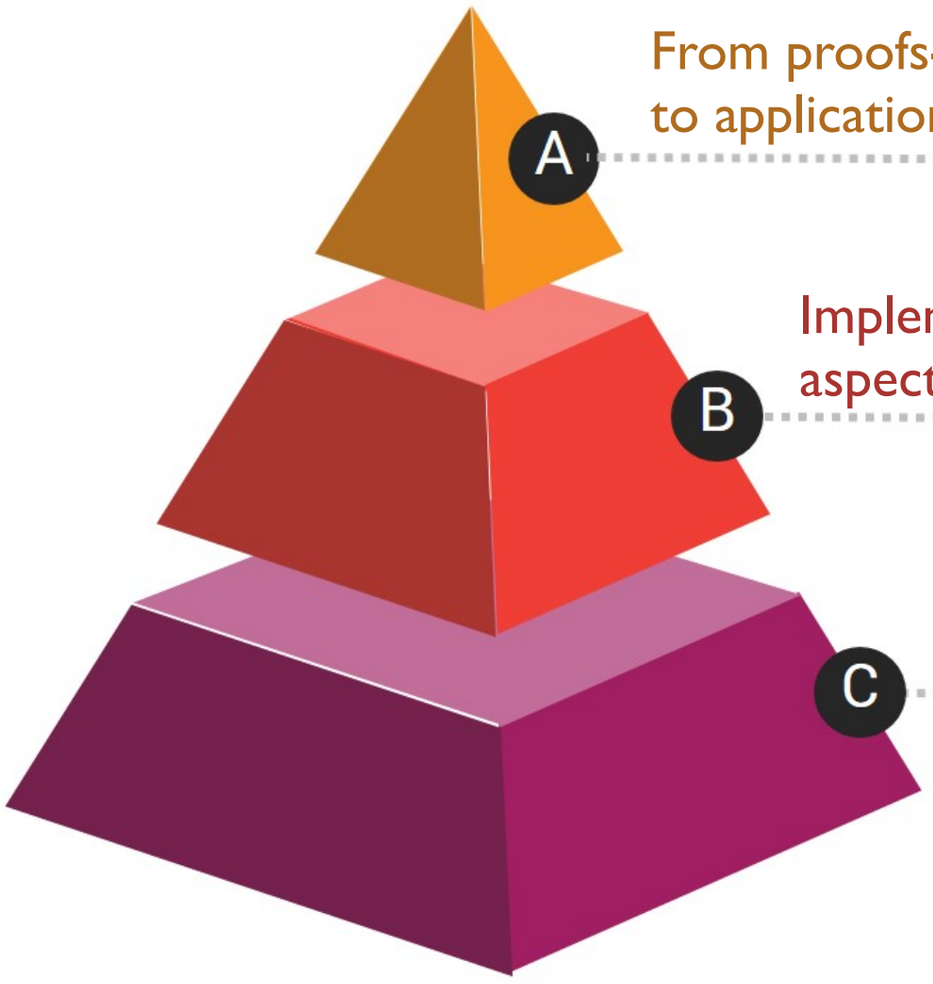








Course philosophy



From proofs-of-concept to applications

Sharpen both the fundamentals as well as your computational skills. Final projects will be geared towards applications.

Implementation aspects

20% of our lecture time will be devoted on hands-on programming tutorials.

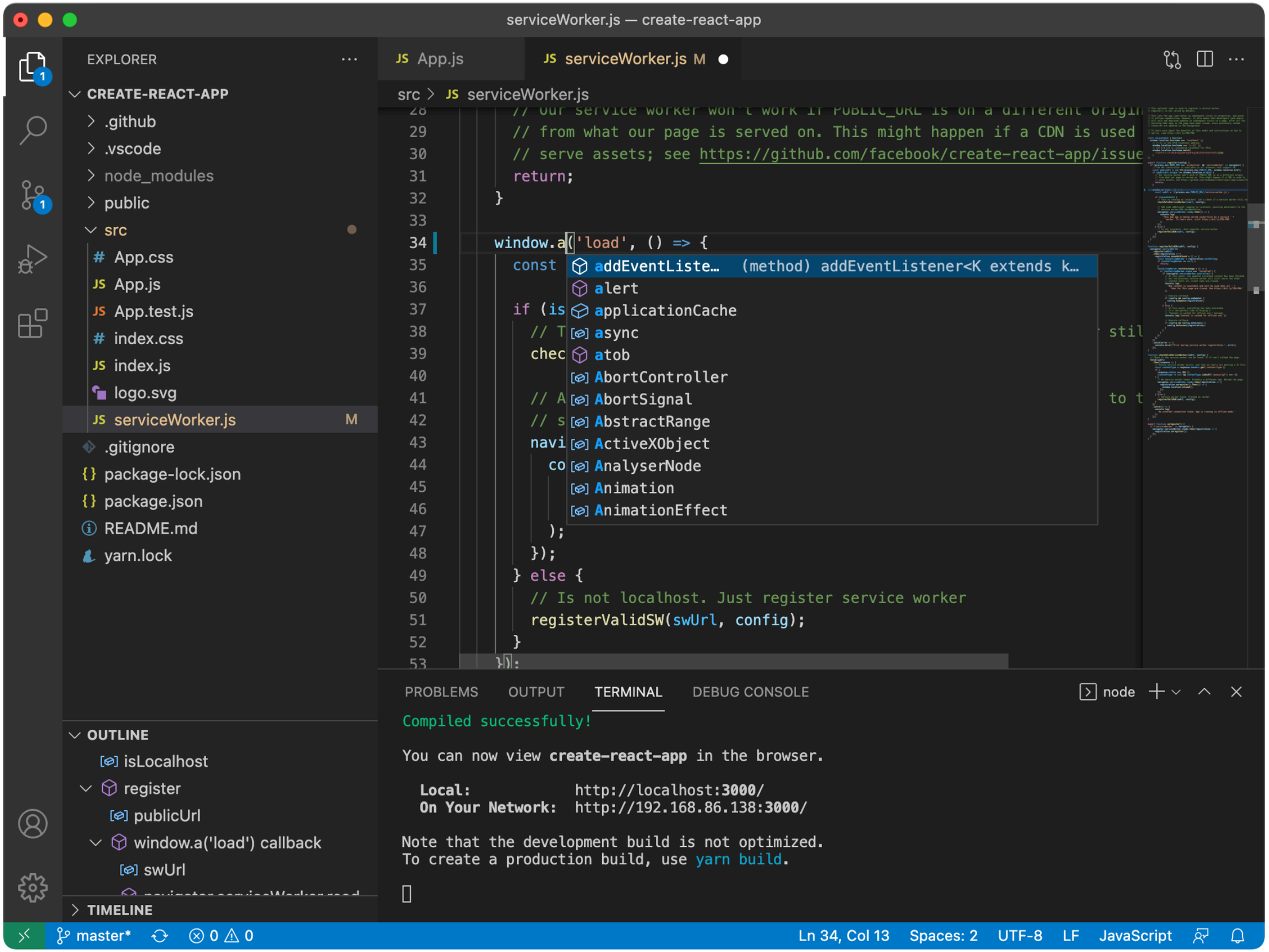
Mathematical foundations

80% of our lecture time will be devoted on discussing the mathematical foundations of modern ML methods.

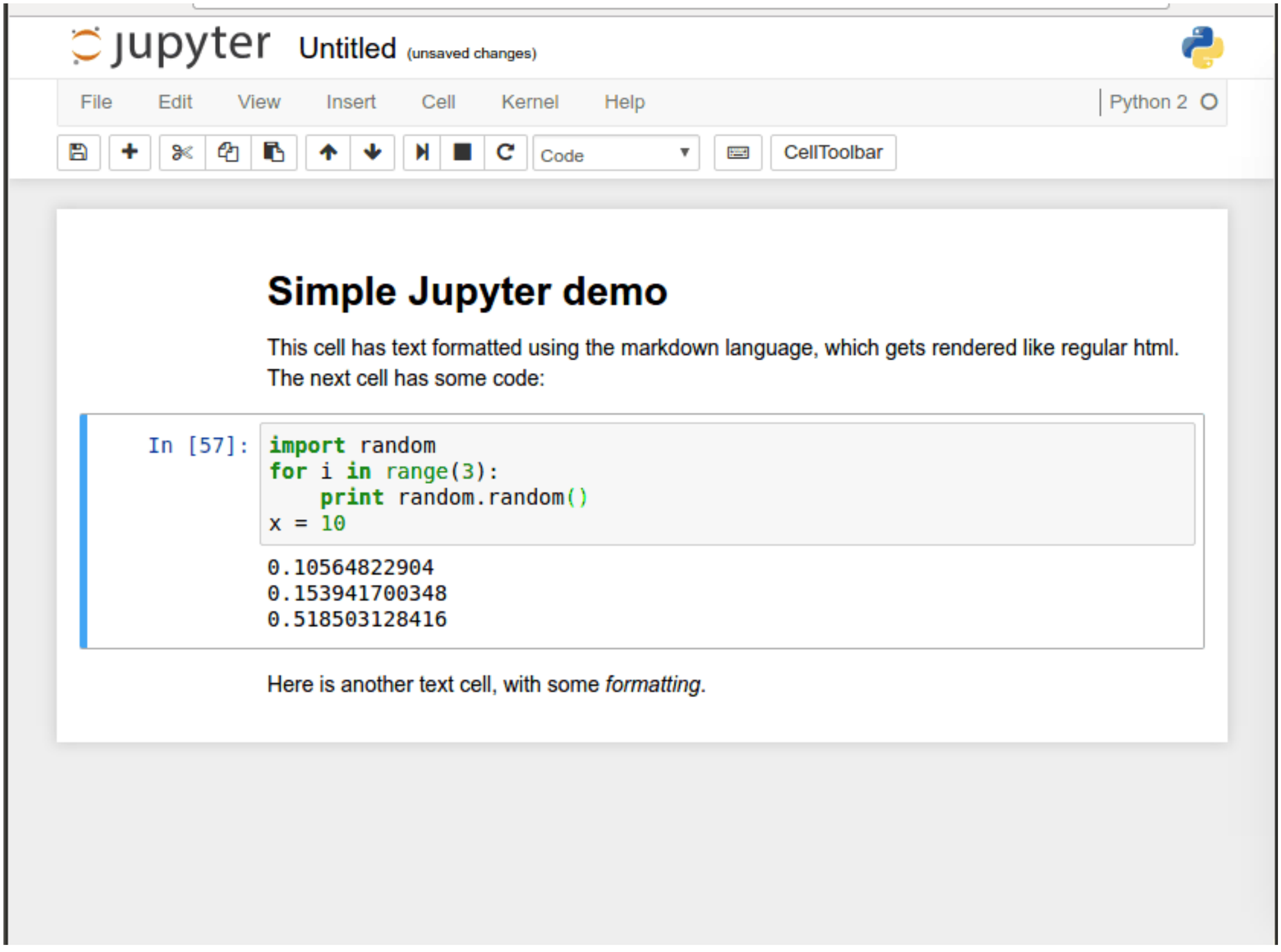
course evaluation plan

Component	Weight	Setting
Practice sets	**ungraded**	your own time
Quizzes	10%	in class, closed book
Coding assignments	30%	open-AI, + oral defense
Midterm	15%	in class, closed book
Final	15%	in class, closed book
Project	30%	open-AI, + presentation

VSCode


serviceWorker.js — create-react-app
EXPLORER
CREATE-REACT-APP
.github
.vscode
node_modules
public
src
App.css
App.js
App.test.js
index.css
index.js
logo.svg
serviceWorker.js M
.gitignore
package-lock.json
package.json
README.md
yarn.lock
OUTLINE
isLocalhost
register
publicUrl
window.a('load') callback
swUrl
TIMELINE
App.js
serviceWorker.js M
src > serviceWorker.js
// from what our page is served on. This might happen if a CDN is used
// serve assets; see https://github.com/facebook/create-react-app/issue
return;
}
window.a('load', () => {
addEventListe… (method) addEventListener<K extends k…
alert
applicationCache
async
atob
AbortController
AbortSignal
AbstractRange
ActiveXObject
AnalyserNode
Animation
AnimationEffect
);
});
} else {
// Is not localhost. Just register service worker
registerValidSW(swUrl, config);
}
PROBLEMS
OUTPUT
TERMINAL
DEBUG CONSOLE
node
Compiled successfully!
You can now view create-react-app in the browser.
Local: http://localhost:3000/
On Your Network: http://192.168.86.138:3000/
Note that the development build is not optimized.
To create a production build, use yarn build.
master*
0 0
Ln 34, Col 13
Spaces: 2
UTF-8
LF
JavaScript

Jupyter notebook



Google colab

face_detection - Google Drive | face_detection.ipynb - Colaborat | colab.research.google.com/drive/1ss4n

Apps | Google | Varzesh3 | P30 | شرکت | PART Services | Paper/Code | trackers | Online courses | MCTS | RaspBerry Pi 4 | Gesture recognition | Cool Tricks | ROAD | Other bookmarks

face_detection.ipynb

File Edit View Insert Runtime Tools Help All changes saved

Comment Share m

Files

Upload Refresh Mount Drive

..

sample_data

+ Code + Text

RAM Disk Editing

Using Google Colab to run our machine learning projects.

Change the directory to face_detection folder

First create a folder and name it as **face_detection**

```
from google.colab import drive
drive.mount('/content/drive/')
```

use `os.chdir` to connect to the face_detection directory

```
[ ] from os import chdir
    chdir('/content/drive/My Drive/face_detection')
```

Clone your project in the google drive

```
[ ] !git clone https://github.com/masouduut94/Pytorch_Retinaface.git
```

```
Cloning into 'Pytorch_Retinaface'...
remote: Enumerating objects: 26, done.
remote: Counting objects: 100% (26/26), done.
remote: Compressing objects: 100% (18/18), done.
remote: Total 146 (delta 7), reused 23 (delta 5), pack-reused 120
Receiving objects: 100% (146/146), 12.34 MiB | 24.73 MiB/s, done.
Resolving deltas: 100% (46/46), done.
```

Doing a little bit cleaning.

```
[ ] !mv Pytorch_Retinaface/* ./
    !rm -r Pytorch_Retinaface/
```

Get the pretrained weight files

1 - We open the link to the pretrained models.

2 - We right click on the weight file and choose "Add Shortcut to Drive"

Disk 37.15 GB available

https://colab.research.google.com/notebooks/intro.ipynb